

Generation Interconnection System Impact Study Report

For Queue Project Q16-0656

Chesapeake Beach 138 kV

August 26, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0656. The Interconnection Customer proposes a 132.0 MW solar photovoltaic facility interconnecting at the Chesapeake Beach 138 kV substation in the Maryland area. The request was received on July 28, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 312 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,654,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Byron Bethany – Chalk Pt 138 kV
Contingency: Chalk Point 345 kV/138 kV Autotransformer
Loading: 139.4% (Rating: 245 MVA)
DFAX: 7.0%
Pre-project loading: 135.6%
2. Facility: Byron Bethany – Dvall 138 kV L
Contingency: Chalk Pt 345/138 kV XFMR
Loading: 136.2% (Rating: 234 MVA)
DFAX: 7.0%
Pre-project loading: 132.3%
3. Facility: Aqueduct – Duvall 138 kV
Contingency: Chalk Point 345/138 kV Auto
Loading: 123.5% (Rating: 218 MVA)
DFAX: 7.0%
Pre-project loading: 119.3%
4. Facility: Aqueduct – Oak Grove 138 kV
Contingency: Chalk Pt 345-138 kV Auto
Loading: 120.5% (Rating: 208 MVA)
DFAX: 7.0%
Pre-project loading: 116.0%
5. Facility: Bay Ridge – Dixie Vly 138 kV
Contingency: Riva Rd – Shingletown 345 kV
Loading: 117.5% (Rating: 451 MVA)
DFAX: 16.6%
Pre-project loading: 112.6%
6. Facility: Dixie Vly – Stranger Crk 138 kV
Contingency: Riva Road – Shingletown 345 kV
Loading: 116.4% (Rating: 467 MVA)

DFAX: 16.6%

Pre-project loading: 111.7%

7. Facility: Stranger Creek – University Of Northern Iowa 138 kV line

Contingency: Riva Rd – Shingletown 345 kV

Loading: 115.4% (Rating: 484 MVA)

DFAX: 16.6%

Pre-project loading: 110.9%

8. Facility: Rush Substation – Colesville 138 kV Line

Contingency: Taber – Glenannan 345 kV L

Loading: 105.5% (Rating: 503 MVA)

DFAX: 9.0%

Pre-project loading: 103.1%

9. Facility: Montpelier – Prairie Wind Farm 138 kV

Contingency: Taber – Glenannan 345 kV

Loading: 100.7% (Rating: 587 MVA)

DFAX: 7.2%

Pre-project loading: 99.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Byron Bethany – Chalk Pt 138 kV	Rebuild line with higher-capacity conductor	\$18,368,000	17% (\$3,062,000)
Byron Bethany – Dvall 138 kV L	Rebuild line with higher-capacity conductor	\$16,831,000	17% (\$2,805,000)
Aqueduct – Duvall 138 kV	Rebuild line with higher-capacity conductor	\$3,812,000	17% (\$635,000)
Aqueduct – Oak Grove 138 kV	Rebuild line with higher-capacity conductor	\$2,151,000	17% (\$358,000)
Bay Ridge – Dixie Vly 138 kV	Rebuild line with higher-capacity conductor	\$6,657,000	17% (\$1,117,000)
Dixie Vly – Stranger Crk 138 kV	Rebuild line with higher-capacity conductor	\$1,405,000	17% (\$236,000)
Stranger Crk – University Of Northern Iowa 138 kV L	Rebuild line with higher-capacity conductor	\$1,789,000	17% (\$300,000)
Rush Substation – Colesville 138 kV Line	Reconductor line	\$1,470,000	4% (\$58,000)
Prairie Wind Farm – Montpelier 138 kV L	Reconductor line	\$1,324,000	6% (\$83,000)
	Total allocated to project		\$8,654,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.